

Erdős Problem #430

EP-430

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Fix some integer n and define a decreasing sequence in $[1, n)$ by $a_1 = n - 1$ and, for $k \geq 2$, letting a_k be the greatest integer in $[1, a_{k-1})$ such that all of the prime factors of a_k are $> n - a_k$.

Is it true that, for sufficiently large n , not all of this sequence can be prime?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>